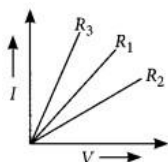


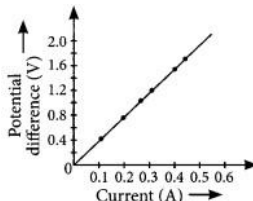
PHYSICS CLASS-X

ELECTRICITY

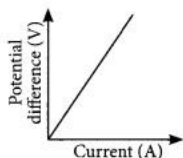
1. A current of 10 A flows through a conductor for two minutes.
(i) Calculate the amount of charge passed through any area of cross section of the conductor.
(ii) If the charge of an electron is 1.6×10^{-19} C, then calculate the total number of electrons flowing.
2. Name a device that you can use to maintain a potential difference between the ends of a conductor. Explain the process by which this device does so.
3. A student plots I-V graphs for three samples of nichrome wire with resistances R_1 , R_2 and R_3 . Choose from the following the statements that holds true for this graph.



- (a) $R_1 = R_2 = R_3$ (b) $R_1 > R_2 > R_3$ (c) $R_3 > R_2 > R_1$ (d) $R_2 > R_1 > R_3$
4. A V-I graph for a nichrome wire is given below. What do you infer from this graph? Draw a labelled circuit diagram to obtain such a graph.



5. V-I graph for a conductor is as shown in the figure



- (i) What do you infer from this graph? (ii) State the law expressed here.
6. State Ohm's law. Draw a labeled circuit diagram to verify this law in the laboratory. If you draw a graph between the potential difference and current flowing through a metallic conductor, what kind of curve will you get? Explain how would you use this graph to determine the resistance of the conductor.
 7. A cylindrical conductor of length 'l' and uniform area of cross section 'A' has resistance 'R'. The area of cross section of another conductor of same material and same resistance but of length '2l' is (2020)
(a) $A/2$ (b) $3A/2$ (c) $2A$ (d) $3A$
 8. (i) List three factors on which the resistance of a conductor depends.
(ii) Write the SI unit of resistivity.
 9. (a) List the factors on which the resistance of a conductor in the shape of a wire depends.
(b) Why are metals good conductors of electricity whereas glass is a bad conductor of electricity? Give reason.
(c) Why are alloys commonly used in electrical heating device.

2. Calculate the number of electrons constituting one coulomb of charge per second through any cross section of a conductor.
3. How much work is done in moving a charge of 2 C across two points having a potential difference of 12 V ?
4. How much energy is given to each coulomb of charge passing through a 6 V battery ?
5. Calculate the work done in moving a charge of 5 Coulombs from a point at a potential of 210 Volts to another point at 240 Volts.
7. When a particle of charge $10\ \mu\text{C}$ is brought from infinity to a point P, 2.0 mJ work is done by the external forces. What is the potential at P?
9. How many electrons pass through a lamp in one minute if the current be 200 mA?
10. Calculate the current supplied by a cell if the amount of charge passing through the cell in 4 seconds is 12 C ?
11. A 2 Volt cell is connected to a $1\ \Omega$ resistor. How many electrons come out of the negative terminal of the cell in 2 minutes ?
12. a) How much current will an electric bulb draw from a 220 V source, if the resistance of the bulb filament is $1200\ \Omega$?
b) How much current will an electric heater coil draw from a 220 V source, if the resistance of the heater coil is $100\ \Omega$?
13. The potential difference between the terminals of an electric heater is 60 Volts when it draws a current of 4 Ampere from a source. What current will the heater draw if the potential difference is increased to 120 Volt.
14. A torch bulb when cold has $1\ \Omega$ resistance. It draws a current of 0.3 Ampere when glowing from a source of 3 V. Calculate the resistance of the bulb when glowing and explain the reason for the difference in resistance.
15. A current of 0.2 Ampere flows through a conductor of resistance $4.5\ \Omega$. Calculate the potential difference at the ends of the conductor.
10. Calculate the resistivity of the material of a wire of length 1 m, radius 0.01 cm and resistance 20 ohms.
11. A wire has a resistance of $16\ \Omega$. It is melted and drawn into a wire of half its original length. Calculate the resistance of the new wire. What is the percentage change in its resistance?
12. If the radius of a current carrying conductor is halved, how does current through it change?